Adrenal Disorders II: Adrenal Insufficiency, Addisonian Crisis, CAH, Pheochromocytoma, Hypogonadism & Hirsutism

Internal Medicine Department | Source: `13-Adrenal insufficiency – CAH – Pheochromocytoma.pdf` (52 slides)

> **Legend:** ⭐ = high-yield · *[added]* = not on slides. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes.
>
> 🔗 **Bridge from earlier lectures:**
>
> - **Adrenal I:** zones and the steroid pathway (21-, 11β-, 17α-hydroxylase). Too much cortisol was Cushing's; here it is **too little**.
> - **Pituitary II: Simmonds disease vs Addison's** (pale vs pigmented, normal vs high K^+) and the rule **steroid before thyroxine**.
> - **Hypertension:** pheochromocytoma's **5 P's** and metanephrines; **α before β**.
> - **Hyperparathyroidism / Thyroid 2: MEN 2A** (medullary thyroid cancer + pheo + parathyroid). This lecture's case.
> - **Pituitary I:** prolactin and Klinefelter (hypogonadism).

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1. Title / Objectives

Domain	Objective	Likely tested as
Knowledge	Aetiology of adrenal disorders	Autoimmune 80%, TB, Waterhouse–Friderichsen
Knowledge	⭐ Manifestations and management of **hypo**adrenalism	Pigmentation, ↓Na ↑K, ACTH stimulation test
Knowledge	Causes and manifestations of **hirsutism**	PCOS vs CAH vs tumour
Skill	⭐ Interpret investigations	1ry vs 2ry vs 3ry AI; CAH enzyme from BP and K^+
Skill	⭐ Emergency management	**Adrenal crisis: hydrocortisone 100 mg IV + saline + dextrose immediately**
Skill	⭐ Pheochromocytoma management	**α-blockade before β-blockade**

Main subtopics: Addison's (causes, picture, investigations, DDx) · adrenal crisis · CAH (17α, 21, 11β) · pheochromocytoma · male hypogonadism · hirsutism.

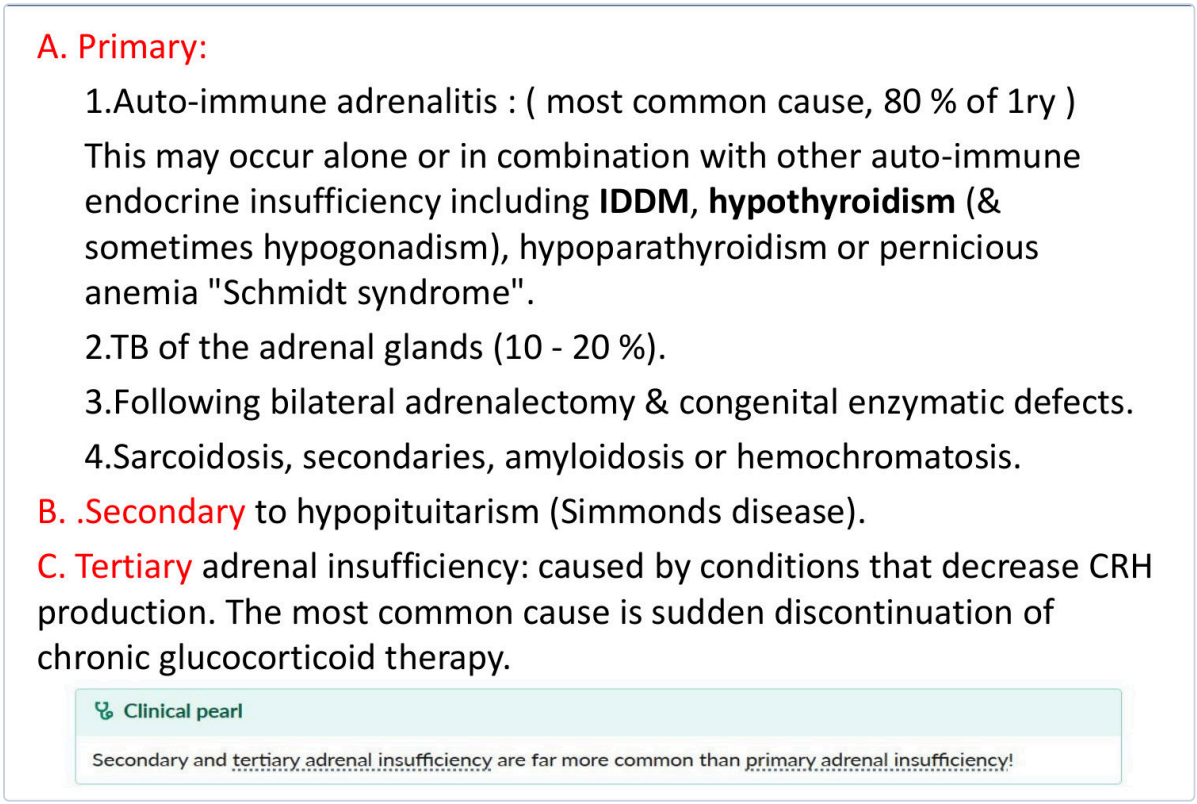
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2. Main Content (Lecture Order)

2.1 Addison's Disease (Chronic Adrenal Failure): Causes (Slide 5) ⭐

Type	Causes
A. Primary (adrenal)	⭐ **Autoimmune adrenalitis: 80% of primary**. Alone or with other autoimmune endocrinopathies (**T1DM, hypothyroidism**, hypogonadism, hypoparathyroidism, pernicious anaemia) = ⭐ **Schmidt syndrome** (*[added]* APS-2) · ⭐ **TB of the adrenals: 10–20%** · **bilateral adrenalectomy**, congenital enzyme defects · **sarcoidosis, metastases, amyloidosis, haemochromatosis**
B. Secondary	**Hypopituitarism** (Simmonds disease)
C. Tertiary	↓ **CRH**. ⭐ **Commonest cause: sudden withdrawal of chronic glucocorticoid therapy**

- **Clinical pearl:** secondary and tertiary AI are **far more common** than primary.
- *[added]* Other primary causes: adrenal haemorrhage (anticoagulants, **Waterhouse–Friderichsen**), HIV/CMV, fungal infection, adrenoleukodystrophy, drugs (ketoconazole).



Slide 5: causes of adrenal insufficiency

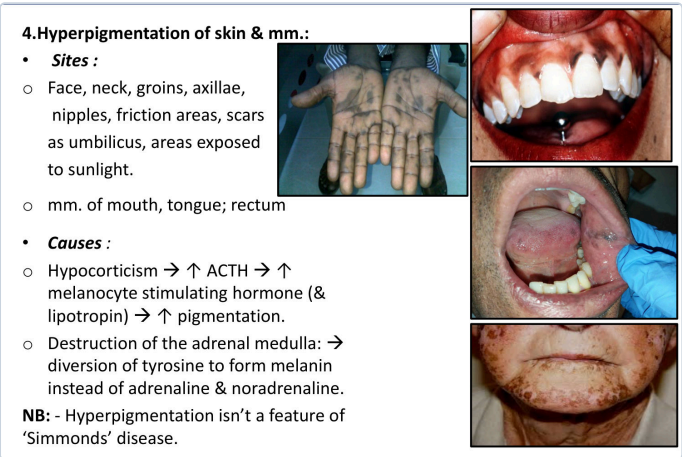
2.2 Clinical Picture (Slides 7–11) ⭐

Onset: usually **insidious**, in adult life; slowly progressive.

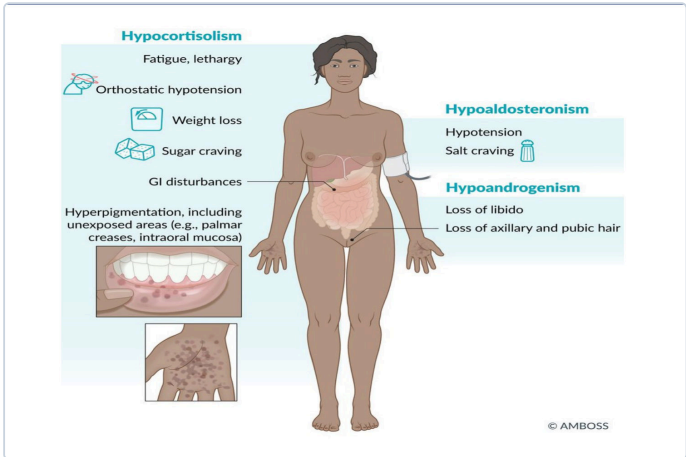
#	Feature	Detail
1	**Asthenia**	**Muscle weakness** (+ **cramps** from hyponatraemia), **weight loss** (lack of cortisol, hyperkalaemia, hypoglycaemia)
2	⭐ **Hypotension**	⭐ **SBP ≥ 110 mmHg (slide: "→ 110") excludes Addison's**. **Postural hypotension** is common (hypovolaemia, dehydration)
3	**Hypoglycaemia**	Drowsiness, hunger pain, even **coma**

#	Feature	Detail
4	⭐ **Hyperpigmentation** (skin and mucous membranes)	**Sites:** face, neck, groins, axillae, **nipples**, **friction areas**, **scars**, umbilicus, sun-exposed areas; **mouth, tongue, rectum** (buccal mucosa). **Mechanism:** ↓ cortisol → **↑ ACTH → ↑ MSH (and lipotropin)**; destruction of the medulla diverts tyrosine to **melanin**. ⭐ **NOT a feature of Simmonds disease**
5	**Loss of adrenal androgen**	**↓ axillary and pubic hair in WOMEN** (men keep testicular androgen); **infertility and amenorrhoea**
6	**GIT**	Nausea, vomiting, **diarrhoea** (even steatorrhoea)
7	**Polyuria**	**Na diuresis**. Renal-angle tenderness in adrenal TB
—	**Addisonian crisis**	May be the first presentation

Clinical pearl: signs of aldosterone deficiency (salt craving, hyperkalaemia, hypotension) separate **primary** from secondary/tertiary.



Slide 8: hyperpigmentation (palmar creases, buccal mucosa, gums, face)



Slide 10: features by hormone (hypocortisolism, hypoaldosteronism, hypoandrogenism)

Slide 11: features by hormone and level ⭐

Hormone loss	Clinical	Labs	1ry	2ry	3ry
Hypoaldosteronism	Hypotension, **salt craving**	**↓Na, ↑K, normal anion gap metabolic acidosis**	✓	**Absent**	**Absent**
Hypocortisolism	Weight loss, anorexia, fatigue, depression, myalgia, weakness, **GI symptoms**, **sugar craving**, orthostatic hypotension	**Hypoglycaemia, hyponatraemia**	✓	✓	✓
Hypoandrogenism	↓ libido, ↓ axillary and pubic hair	**↓ DHEA-S**	✓	✓	✓

Hormonal changes	Clinical features	Laboratory findings	Primary adrenal insufficiency	Secondary adrenal insufficiency	Tertiary adrenal insufficiency
Hypoaldosteronism	• Hypotension • Salt craving	• Hyponatremia • Hyperkalemia • Normal anion gap metabolic acidosis	✓	• Absent	• Absent
Hypocortisolism	• Weight loss, anorexia • Fatigue, lethargy, depression • Muscle aches • Weakness • Gastrointestinal complaints (e.g., nausea, vomiting, diarrhea) • Sugar cravings • (Orthostatic) hypotension	• Hypoglycemia • Hyponatremia	✓	✓	✓
Hypoandrogenism	• Loss of libido • Loss of axillary and pubic hair	• ↓ DHEA-S	✓	✓	✓

Slide 11: hormonal changes by level of adrenal insufficiency

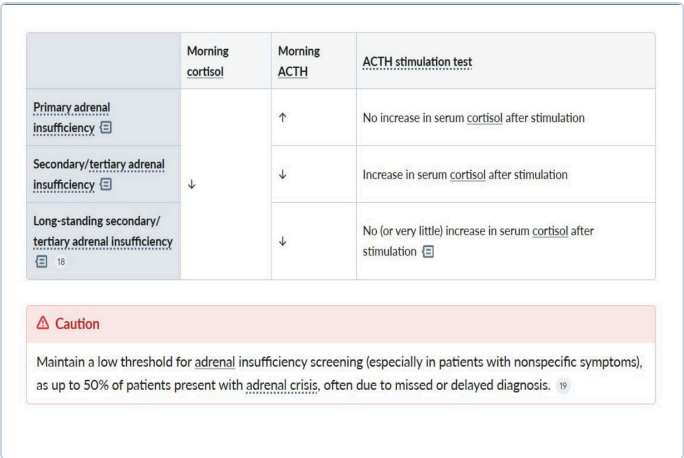
2.3 Investigations (Slides 12–15) ⭐

#	Test	Finding
1	**Plasma cortisol**	↓ (morning)
2	⭐ **Morning ACTH** (if cortisol low)	**Primary → ↑ ACTH**; **secondary/tertiary → low or normal**
3	**24-h urine**	↓ cortisol, 17-OHCS, 17-ketosteroids
4	⭐ **ACTH (cosyntropin) stimulation test: GOLD STANDARD**	**Normal: cortisol ↑ ≥ 6 µg/dL above basal** (*[added]* modern: peak ≥ 18 µg/dL). **Primary → no response**. **Secondary → normal response**. **Tertiary → none or mild** (adrenal atrophy from lack of stimulation)
5	**Biochemistry**	⭐ **↓ Na, ↓ Cl; ↑ K; acidosis; ↑ Ca; ↓ glucose**
6	**Urine**	↑ volume; **↑ Na and Cl, ↓ K excretion**
7	**Blood picture**	**Normochromic anaemia, haemoconcentration, eosinophilia, lymphocytosis**
8	**Cause**	**Adrenal antibodies** (*[added]* 21-hydroxylase Ab) → autoimmune; **X-ray: adrenal calcification** (TB), pulmonary TB

Slide 14 summary (and caution):

	Morning cortisol	Morning ACTH	ACTH stimulation
Primary	↓	↑	**No increase**
Secondary/tertiary	↓	↓	**Increase**
Long-standing secondary/tertiary	↓	↓	**No or very little increase** (atrophy)

⚠️ **Caution:** keep a **low threshold for screening**. **Up to 50% of patients present in adrenal crisis**, often from a missed or delayed diagnosis.

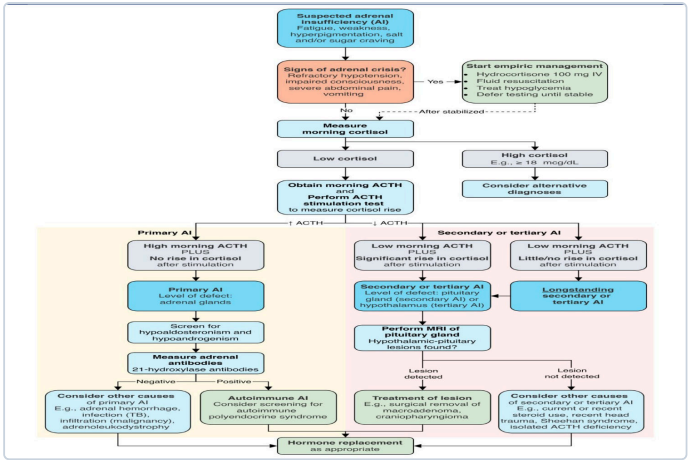


	Morning cortisol	Morning ACTH	ACTH stimulation test
Primary adrenal insufficiency	↓	↑	No increase in serum cortisol after stimulation
Secondary/tertiary adrenal insufficiency		↓	Increase in serum cortisol after stimulation
Long-standing secondary/ tertiary adrenal insufficiency 18		↓	No (or very little) increase in serum cortisol after stimulation

⚠ Caution

Maintain a low threshold for adrenal insufficiency screening (especially in patients with nonspecific symptoms), as up to 50% of patients present with adrenal crisis, often due to missed or delayed diagnosis. 19

Slide 14: cortisol / ACTH / stimulation test by level



Slide 15: adrenal insufficiency diagnostic algorithm

Algorithm (slide 15):

1. Suspected AI: **signs of crisis? → empiric hydrocortisone 100 mg IV, fluids, treat hypoglycaemia; defer testing until stable**.
2. Otherwise measure **morning cortisol**: - **high (≥ 18 µg/dL)** → consider alternative diagnoses - **low → morning ACTH + ACTH stimulation test**
3. Interpret: - **↑ ACTH + no rise = primary** → screen for hypoaldosteronism and hypoandrogenism → **adrenal (21-hydroxylase) antibodies**: - positive → autoimmune; screen for **polyendocrine syndrome** - negative → haemorrhage, infection (TB), malignancy, adrenoleukodystrophy - **↓ ACTH + rise = secondary/tertiary → pituitary MRI**: - lesion → treat it (macroadenoma, craniopharyngioma) - none → recent steroid use, head trauma, **Sheehan**, isolated ACTH deficiency

2.4 Differential Diagnosis (Slide 16)

1. **Secondary hypoadrenalism** (no pigmentation, normal K^+)
2. **Other causes of pigmentation:** racial, **haemochromatosis**, porphyria, acanthosis nigricans, neurofibromatosis
3. **Asthenia + pigmentation: pellagra**, **CML** (especially on **busulfan**), **CRF**, chronic **arsenic** poisoning, thyrotoxicosis, advanced malignancy, malabsorption

2.5 Treatment (Slide 17) ⭐

A. Replacement:

- **Glucocorticoid: prednisolone 5 mg morning + 2.5 mg evening** (*[added]* or hydrocortisone 15–25 mg/day in 2–3 doses)
- ⭐ **Mineralocorticoid: fludrocortisone 0.1 mg** (slide: "every other day"; *[added]* usually daily), **primary only**
- **Active TB → anti-TB therapy**

B. Stress dosing ("sick-day rules"):

Situation	Action
Minor illness (dental extraction, URTI)	⭐ **Double the dose** until resolved
Elective major surgery	**Hydrocortisone 100 mg IV the night before**, then **100 mg every 8 h**; taper over **3–5 days** to the usual dose
Catastrophes / emergencies (trauma, MI, major emergency surgery)	**Treat as acute adrenal failure**

A. Replacement therapy :

- Glucocorticoids →prednisolone 5 mg in the morning & 2.5 mg in the evening.
- Mineralocorticoids → fludrocortisone 0.1 mg every other day.

N.B: Active TB. cases should receive anti-TB therapy.

B. Management during stressful conditions :

Minor illness	Major illness	
e.g. - Dental extraction. - Upper respiratory tract infection • The dose is doubled "until the condition is resolved".	For elective major surgery	For catastrophies & emergencies
	• 100 mg hydrocortisone IV during the night preceedmg surgery & then 100 mg / 8 hours. • Then →the dose is gradually tapered within 3 — 5 days to reach the previous dose.	e.g. - trauma, - myocardial infarction - major emergency surgery, …etc → ttt as acute adrenal failure.

Slide 17: replacement and stress dosing

🔁 **Recap: Chronic AI.** Primary: autoimmune 80% (Schmidt), TB 10–20%, metastases, haemorrhage. Secondary: pituitary. Tertiary: steroid withdrawal (commonest overall). Features: weakness, weight loss, postural hypotension (SBP ≥ 110 excludes), hypoglycaemia, **pigmentation (primary only)**, ↓ female body hair, GI upset, salt craving. Labs: ↓Na ↑K acidosis ↑Ca ↓glucose, eosinophilia. ACTH stimulation (gold standard): primary no rise + high ACTH; secondary rise + low ACTH. Treatment: prednisolone 5 + 2.5 mg + fludrocortisone 0.1 mg (primary); double the dose for minor illness, IV hydrocortisone for surgery.

🔗 **Link:** Untreated or stressed, chronic insufficiency decompensates within hours.

▶️ **Up next: Adrenal crisis.**

2.6 Acute Adrenal Failure: Addisonian Crisis (Slides 18–21) ⭐

Causes:

1. **Undiagnosed patient** presenting spontaneously or with intercurrent illness
2. ⭐ Patient on cortisone facing severe illness **without increasing the dose**
3. ⭐ **Sheehan syndrome treated with thyroxine alone, without cortisone**
4. **After bilateral adrenalectomy** or pituitary surgery
5. ⭐ **Sudden corticosteroid withdrawal**
6. ⭐ **Acute meningococcal septicaemia → adrenal haemorrhage** = **Waterhouse–Friderichsen syndrome**
7. **Medical therapy for adrenal carcinoma** (metyrapone, mitotane)
8. **Massive adrenal vein thrombosis** (pregnancy, puerperium, burns)

Clinical picture:

- **Sudden** severe weakness, **confusion**, ⭐ **abdominal pain**, anorexia, nausea, **vomiting**, diarrhoea
- **Dehydration → hypotension → shock**

- **Untreated → coma and death within 24 h**

Caution: most adrenal insufficiency is subclinical and only shows during **stress** (surgery, trauma, infection), when the cortisol requirement rises. ⭐ **Consider adrenal crisis in any patient with hypotension refractory to fluids and/or vasopressors.**

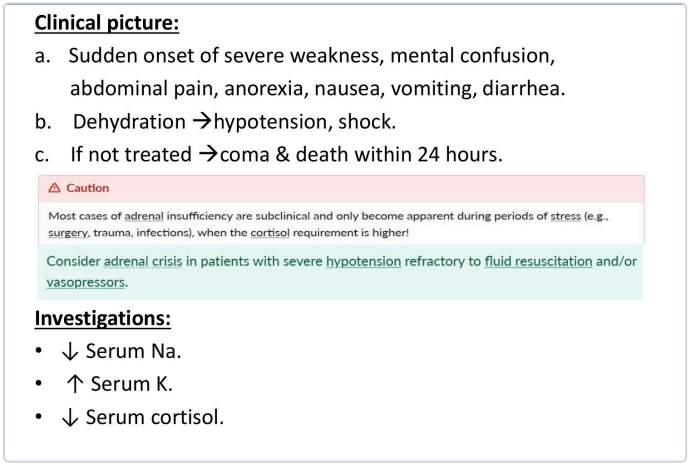
Investigations: ↓ Na, ↑ K, ↓ cortisol.

Treatment (slide 21) ⭐

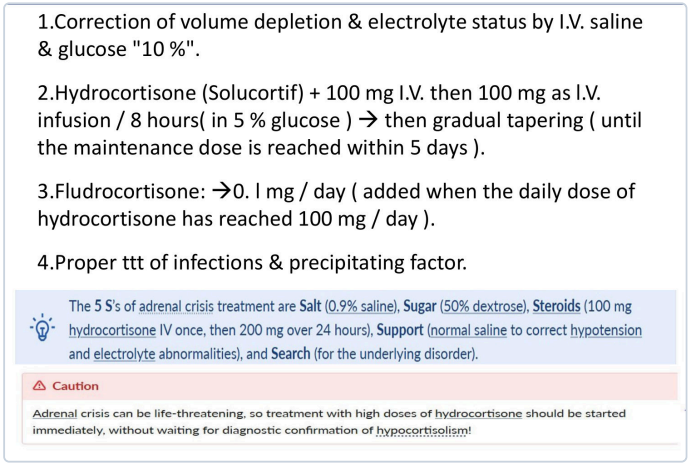
1. **Correct volume and electrolytes**: **IV saline + glucose 10%**
2. ⭐ **Hydrocortisone (Solu-Cortef) 100 mg IV**, then **100 mg IV infusion every 8 h** (in 5% glucose), tapering to the maintenance dose **within 5 days**
3. **Fludrocortisone 0.1 mg/day**, added when hydrocortisone has fallen to **100 mg/day** (*[added]* high-dose hydrocortisone has enough mineralocorticoid effect)
4. **Treat infection** and other precipitating factors

> ⭐ **The 5 S's (slide 21): S**alt (0.9% saline) · **S**ugar (50% dextrose) · **S**teroids (**hydrocortisone 100 mg IV once, then 200 mg over 24 h**) · **S**upport (saline for hypotension and electrolytes) · **S**earch (for the underlying cause).

⚠️ **Caution:** adrenal crisis is life-threatening. **Start high-dose hydrocortisone IMMEDIATELY, without waiting for diagnostic confirmation.** *[added] Draw cortisol and ACTH first if it doesn't delay treatment. If testing is needed later, use dexamethasone, because it doesn't cross-react with the cortisol assay.*



Slide 20: crisis clinical picture and caution

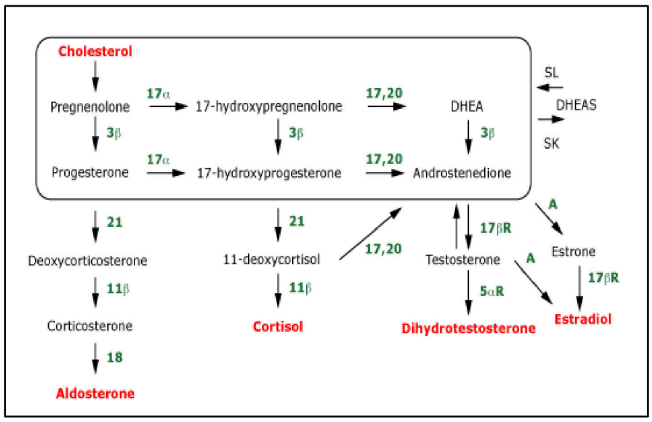


Slide 21: crisis treatment and the 5 S's

> 🔁 **Recap: Crisis.** Triggers: illness without stress dosing, steroid withdrawal, thyroxine before steroid in Sheehan, adrenalectomy, meningococcaemia (Waterhouse–Friderichsen), adrenal vein thrombosis. Shock, abdominal pain, vomiting, confusion → death within 24 h. Give hydrocortisone 100 mg IV now (no waiting), then 100 mg every 8 h; IV saline + dextrose; fludrocortisone when hydrocortisone ≤ 100 mg/day; treat the trigger.
>
> 🔗 **Link:** Some patients are born unable to make cortisol because of a single missing enzyme.
>
> ▶️ **Up next: Congenital adrenal hyperplasia.**

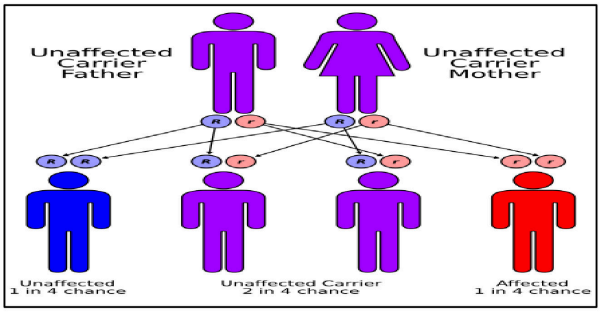
2.7 Congenital Adrenal Hyperplasia (Adrenogenital Syndrome) (Slides 22–26) ⭐

Definition: inherited (*[added]* **autosomal recessive**, slide 24 pedigree: 1 in 4 affected, 2 in 4 carriers) enzyme defects in cortisol synthesis → **↓ cortisol → ↑ ACTH → adrenal hyperplasia + ↑ synthesis of the hormones proximal to the block** (shunted to androgens).



Slide 23: steroid synthesis pathway

- Group of conditions caused by inherited enzymatic defects in the enzymes necessary for the synthesis of cortisol → ↓ cortisol → ↑ ACTH adrenal hyperplasia & ↑ synthesis of the hormone proximal to the block.



Slide 24: CAH definition and autosomal recessive inheritance

Enzyme ↓	Frequency	Hormonal changes	⭐ Clinical features
⭐ **21-hydroxylase**	⭐ **Commonest form** (*[added]* ~95%)	**↓ glucocorticoids AND mineralocorticoids**; **↑ sex steroids**	⭐ **Salt-losing syndrome** (↓ aldosterone): **hypotension, hyperkalaemia** · **Androgen excess:** girls → **pseudohermaphroditism** (big clitoris, small vagina, fused enlarged labia mimicking a scrotum, **normal internal genitalia**); boys → **precocious puberty** (macrogenitosomia praecox) **without spermatogenesis** (small testes), masculinisation, deep voice, **premature epiphyseal fusion → dwarfism**
11β-hydroxylase	**~10%**	**↑ 11-deoxycorticosterone** (a mineralocorticoid), 11-deoxycortisol and **androgens**; ↓ cortisol	⭐ **Hypertension** (+ hypokalaemia *[added]*) · **virilisation** → female pseudohermaphroditism · **precocious puberty**
17α-hydroxylase	**Extremely rare**	**↓ glucocorticoids AND sex hormones**; **↑ mineralocorticoids**	⭐ **HTN + hypokalaemia** · females: **sexual infantilism** (no puberty) · males: **pseudohermaphroditism** (undervirilised)

Mnemonic *[added]*: Look at the number.

- **"1" first (17, 11) → HTN** (mineralocorticoid builds up).
- **"1" second (21, 11) → virilisation** (androgens build up).
- **11β** has both → **HTN + virilisation**.
- **21** → **hypotension + virilisation** (salt-wasting).
- **17** → **HTN + no androgens** (infantilism / undervirilised males).

Enzyme ↓	Hormonal changes	Clinical features
17 oH ase ↓ extremely rare	-↓ glucocorticoids & sex hormones. - ↑ mineralocorticoids	• **HPN & Hypokalemia** • In ♀→ **sexual infantilism** • In ♂→ ♂ **pseudohermaphroditism**
21 oH ase ↓ it is the commonest form	- ↓ glucorticoids & mineralocorticoids. - ↑ sex steroids	• Salt losing syndrome: due to aldosterone ↓: **o Hypotension** **o Hyperkalemia** • Androgen excess: o ♀ **pseudohermaphroditism**: (big clitoris, small vagina, the libia majora are big & fused simulating scrotum+normal internal jenitalia) o ♂ **precocious puberty:** (Macrojenitosmia precox), but without spermatogenesis & the testicle remain small, masculinization, deeping of voice & premature fusion of epiphysis dwarfism
11 B oH ase ↓ about 10% of cases	- ↑ 11 deoxycorticosterone, deoxycortsol & androgns. - ↓ cortisol	• **Hypertension** • Virilization → ♀ **pseudohermaphroditism** • **Precocious puberty**

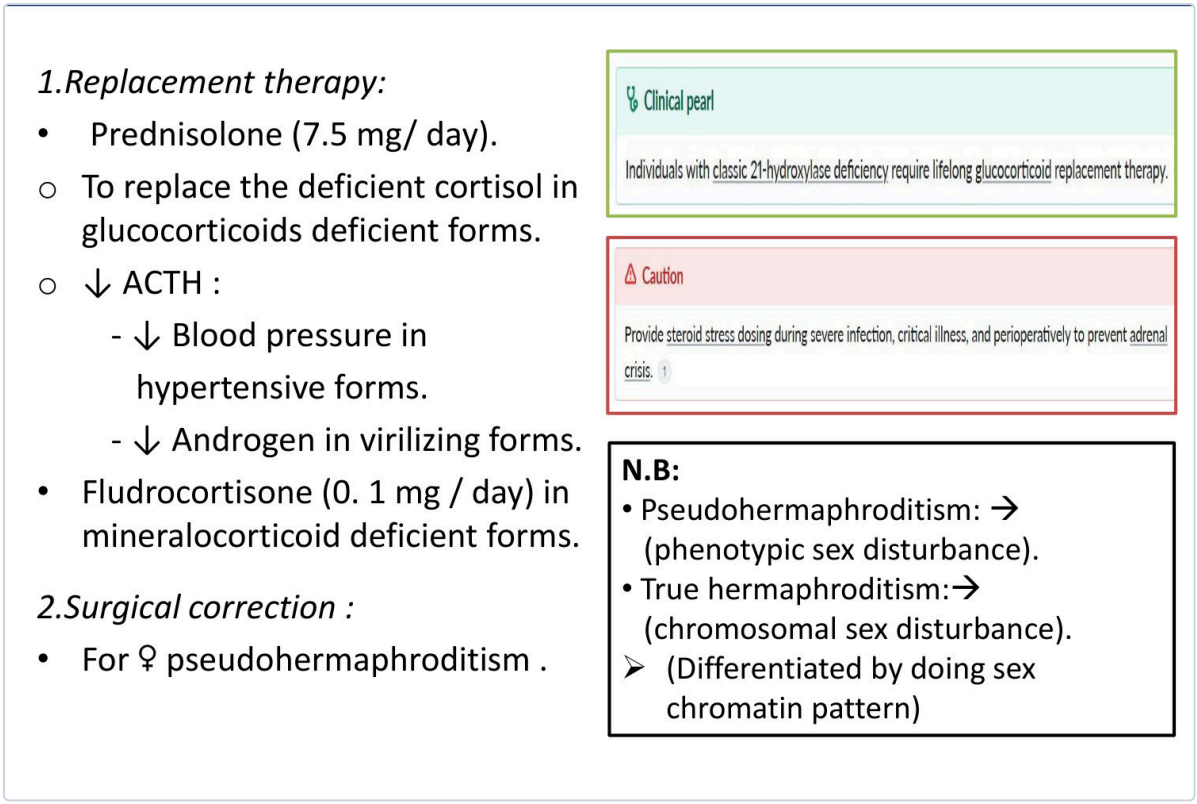
Slide 25: CAH enzyme table

[added] Diagnosis of 21-OH deficiency: ⭐ ↑ 17-hydroxyprogesterone (neonatal screening). 11β: ↑ 11-deoxycortisol. Salt-wasting crisis in a newborn boy at 1–3 weeks is easily missed (normal genitalia).

Treatment (slide 26):

1. **Replacement:** - **Prednisolone 7.5 mg/day** replaces cortisol and **suppresses ACTH**, which → **↓ BP** (hypertensive forms) and **↓ androgens** (virilising forms). - **Fludrocortisone 0.1 mg/day** in **mineralocorticoid-deficient** (salt-losing) forms.
2. **Surgical correction** of female pseudohermaphroditism.

- **Pseudohermaphroditism** = **phenotypic** sex disturbance; **true hermaphroditism** = **chromosomal** sex disturbance. They are told apart by **sex chromatin** (karyotype).
- **Pearl:** CAH patients **need glucocorticoid replacement for life**, and **stress dosing** during infection, illness or surgery.



Slide 26: CAH treatment

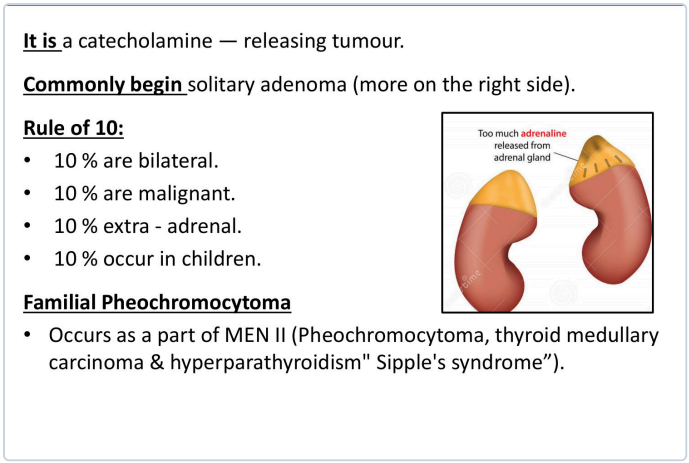
> 🔁 **Recap: CAH.** AR enzyme block → ↓ cortisol → ↑ ACTH → hyperplasia + shunting. **21-OH (commonest):** salt loss (↓BP, ↑K) + virilisation (ambiguous girls, precocious boys, short adults); ↑ 17-OHP. **11β (~10%):** HTN + virilisation (↑ DOC). **17α (rare):** HTN + hypoK + no sex steroids (infantile girls, undervirilised boys). Treatment: prednisolone (suppresses ACTH) ± fludrocortisone; genital surgery; stress dosing.
>
> 🔗 **Link:** Moving from cortex to medulla, the catecholamine tumour.
>
> ▶️ **Up next: Pheochromocytoma.**

2.8 Pheochromocytoma (Slides 27–35) ⭐

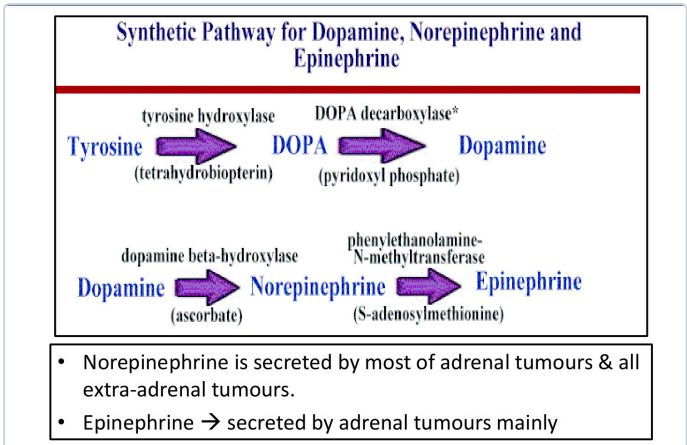
Definition: a **catecholamine-releasing tumour** (*[added]* of chromaffin cells). Commonly a **benign solitary adenoma**, more on the **right**.

> ⭐ **Rule of 10s: 10% bilateral · 10% malignant · 10% extra-adrenal** (*[added]* paraganglioma) **· 10% in children** (*[added]* also 10% familial, 10% normotensive; modern data show ~30–40% hereditary).

- ⭐ **Familial:** part of **MEN 2** (*[added]* 2A and 2B): pheochromocytoma, **medullary thyroid carcinoma**, **hyperparathyroidism** = **Sipple's syndrome** (MEN 2A). *[added] Also VHL, NF1, SDH mutations.*



Slide 28: pheochromocytoma, rule of 10s



Slide 29: catecholamine synthesis

Catecholamine synthesis (slide 29):

1. **Tyrosine** →(tyrosine hydroxylase)
2. **DOPA** →(DOPA decarboxylase)
3. **Dopamine** →(dopamine β-hydroxylase)
4. **Noradrenaline** →(PNMT)
5. **Adrenaline**

- **Noradrenaline** is secreted by **most adrenal** and **all extra-adrenal** tumours.
- **Adrenaline** comes **mainly from adrenal** tumours. *[added]* PNMT needs high local cortisol, which is only present in the adrenal.

Clinical manifestations (slides 30–31):

- ⭐ **Episodic headache, palpitations (rapid heartbeat), sweating, anxiety**, lasting minutes or longer, variable in severity and frequency
- Nausea, vomiting, **tremor**, chest or abdominal pain
- **Cold hands and feet, pale skin, facial pallor** (α-vasoconstriction)
- ⭐ **Hypertension: paroxysmal in 50%, persistent in 50%**; **postural hypotension** can occur
- **Complications** of HTN in the CVS and CNS (*[added]* catecholamine cardiomyopathy, stroke, MI)
- **Constipation, ischaemic enterocolitis**
- **Masses** in the abdomen or neck
- **Metabolic: diabetogenic** (β: ↑ glycogenolysis; α: ↓ insulin secretion)
- ⭐ **Attack triggers:** - **pressure on the tumour** (palpation, micturition *[added]*) - **exercise** - **food** (cheese, alcohol: tyramine) - **drugs**, especially **β-blockers** - **labour**
- **Pearls:** - **5 P's** = **P**ressure (HTN), **P**ain (headache), **P**erspiration, **P**alpitations, **P**allor. - Patients may be **asymptomatic** (incidentaloma).

- Headaches, rapid heartbeat, excessive sweating, and anxiety. These symptoms can occur in episodes that may last for several minutes or longer, and can vary in severity and frequency.
- Other potential symptoms include nausea, vomiting, tremors,, and chest or abdominal pain.
- Cold hands & feet pale skin& pallor of the face. →due to Vasoconstriction (α -stimulation)
- High blood pressure (Hypertension),paroxysmal → in 50 % of cases, while in the other 50 % it is persistent. Postural hypotension may occur.

Clinical pearl

Patients can show typical clinical features (e.g., secondary hypertension) but may also be asymptomatic.

Student pearl

5 most important Problems (5 P's) of Pheochromocytoma: increased blood Pressure, head Pain (headache), Perspiration, Palpitations, and Pallor

Slide 30: clinical manifestations and 5 P's

Investigations (slides 32–33) ⭐

#	Test	Note
1	**24-h urine**: ⭐ **metanephrines** (highly sensitive), **free catecholamines**, **VMA** (N 2–6 mg/day)	**False-positive VMA** from **vanilla, chocolate, aspirin**
2	⭐ **Plasma free metanephrines**	**Highly sensitive** (*[added]* best initial test)
3	**Plasma catecholamines**	
4	**Stimulation tests** (infrequent episodes): **glucagon 1 mg IV** (provokes an attack in 90%), **histamine 50 µg IV**	⚠️ **Phentolamine must be available** (*[added]* largely obsolete, dangerous)
5	⭐ **Clonidine suppression test** (0.3 mg)	Plasma noradrenaline **falls to normal at 3 h in essential HTN**, **not in pheochromocytoma**
6	**Localisation**	**US, CT, MRI**, **radio-isotope scanning** (*[added]* MIBG, Ga-68 DOTATATE PET); **SVC/IVC venous sampling** for extra-adrenal tumours not found radiologically

> ⭐ **Order** *[added]*: **biochemistry FIRST (metanephrines), imaging SECOND.** Never biopsy a suspected pheo (→ crisis).

5.Suppression tests: Clonidine suppression test : (0.3 mg clonidine)

Plasma NA decreases to normal range 3 hours after receiving clonidine in essential HPN, not in Pheochromocytoma.

6.Localization of the tumour :

- o U/S.
- o CT
- o MRI
- o Radio-isotope scanning.
- o Sampling from SVC, IVC :



Determination of plasma catecholamines is useful in extra adrenal tumours that could not be determined by radiological methods.

Slide 33: suppression and localisation tests (CT: adrenal mass)

Treatment (slides 34–35) ⭐

A. Surgical: stabilise BP with sodium nitroprusside or phentolamine → then surgical removal (*[added]* laparoscopic adrenalectomy after 10–14 days of α-blockade + high salt and fluids).

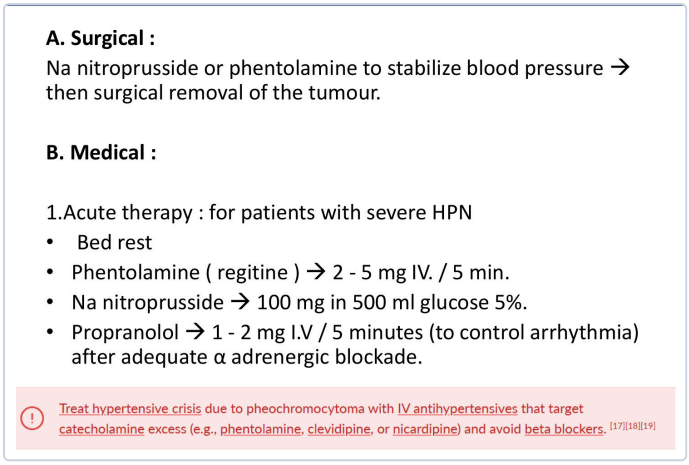
B. Medical:

- **1. Acute therapy (severe HTN / crisis):** - bed rest - ⭐ **phentolamine (Regitine) 2–5 mg IV every 5 min** - **sodium nitroprusside** 100 mg in 500 mL 5% glucose - **propranolol 1–2 mg IV every 5 min** for arrhythmia, **only after adequate α-blockade**
- **2. Prolonged therapy** (e.g., surgery delayed by a recent MI): - ⭐ **phenoxybenzamine (Dibenzyline)**: long-acting **α-blocker**, 10 mg twice daily (↑ by 10–20 mg/day up to 40–200 mg/day) - **propranolol 20–40 mg/day after α-blockade** - **metyrosine**: **tyrosine hydroxylase inhibitor**, pre-operatively or in **inoperable** cases

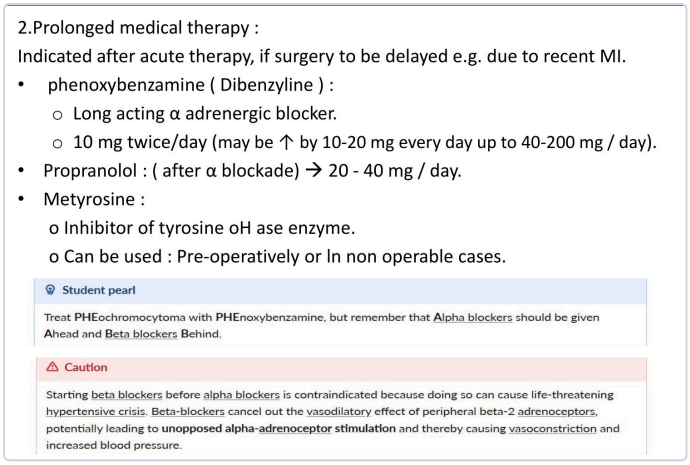
> ⭐ **Student pearl:** treat **PHE**ochromocytoma with **PHE**noxybenzamine, and remember **A**lpha blockers go **A**head, **B**eta blockers **B**ehind.

⚠️ **Caution (slide 35): starting β-blockers before α-blockers is contraindicated.** It blocks peripheral β_2 vasodilatation → **unopposed α-stimulation** → vasoconstriction → **life-threatening hypertensive crisis**.

⚠️ **Slide 34 caution:** treat a hypertensive crisis from pheo with **IV antihypertensives that target catecholamine excess** (phentolamine, clevidipine, nicardipine) and **avoid β-blockers**.



Slide 34: pheochromocytoma treatment (acute)



Slide 35: prolonged therapy; α before β

> 🔁 **Recap: Pheochromocytoma.** Catecholamine tumour; rule of 10s; MEN 2A (Sipple). Episodic headache, sweating, palpitations, pallor; paroxysmal (50%) or sustained HTN; postural drop; hyperglycaemia. Triggers: palpation, exercise, tyramine, β-blockers, labour. Tests: plasma or urine **metanephrines** (most sensitive), VMA (false +ve with vanilla, chocolate, aspirin), clonidine non-suppression; then CT/MRI/MIBG. Treatment: **α first** (phenoxybenzamine; acutely phentolamine/nitroprusside), **β after**, metyrosine, then surgery.

2.9 Male Reproductive Notes: Hypogonadism (Slides 36–38)

- **Leydig cells → testosterone** → feeds back ↓ LHRH → ↓ **LH**.
- **Sertoli cells → inhibin** → ↓ **FSH**; also **Müllerian-inhibiting factor**.
- Most circulating testosterone is bound to **SHBG**.

⭐ Pattern	Meaning
↓ LH, ↓ FSH (↓ testosterone)	**Pituitary/hypothalamic** = **hypogonadotrophic hypogonadism** (*[added]* Kallmann, prolactinoma)
↑ LH, ↑ FSH, ↓ testosterone	⭐ **Primary testicular disease**: irradiation, **myotonic dystrophy**, **cystic fibrosis**, ⭐ **Klinefelter**
Normal/↑ LH, FSH and testosterone	**End-organ (androgen) resistance**
↓ testosterone with NO features of hypogonadism	**↓ SHBG** (free testosterone normal)

Treatment:

- **Androgen replacement IM, not oral** (liver toxicity).
- Pituitary causes → as before (hMG/hCG for fertility).
- **Hypothalamic causes → pulsatile LHRH** (portable pump).

2.10 Hirsutism (Slides 39–47) ⭐

Definitions:

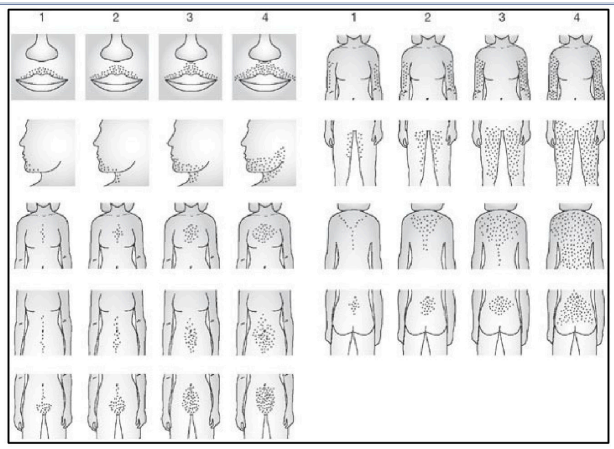
- **Hirsutism** (females only) = ↑ hair in **androgen-dependent, male-pattern** areas: face, chest, lower back, lower abdomen, inner thighs.
- **Hypertrichosis** = excess hair **anywhere on the normal body-hair distribution**, in **either sex** (androgen-independent).

- **Hirsutism** is a term reserved for females and refers to ↑ number of hair in areas associated with ♂ pattern hair growth e.g. face, chest, lower back, lower abdomen, inner thigh.
- **Hypertrichosis** is defined as excessive hair growth anywhere on the normal body hair distribution areas in either males or females.





Slide 40: hirsutism (back and face)



Slide 41: Ferriman–Gallwey scoring diagram

[added] Slide 41 = modified Ferriman–Gallwey score: 9 body areas each scored 0–4; ≥ 8 = hirsutism.

Hair facts (slides 42–43):

- The number of follicles is **set at birth**; hair grows from pilosebaceous units continuously in a **mosaic** pattern (asynchronous).
- Synchronised cycles → transient alopecia or excess.
- **Types:** - **lanugo** (fetal) - **vellus** (fine adult body hair) - **terminal** (thick, pigmented: scalp, pubic)
- **Androgen-sensitive** areas: face, neck, chest, abdomen, axillae, upper arms, inner thighs, pubic hair, part of the scalp.
- **Androgen-independent** areas: forearms, hands, lower legs.

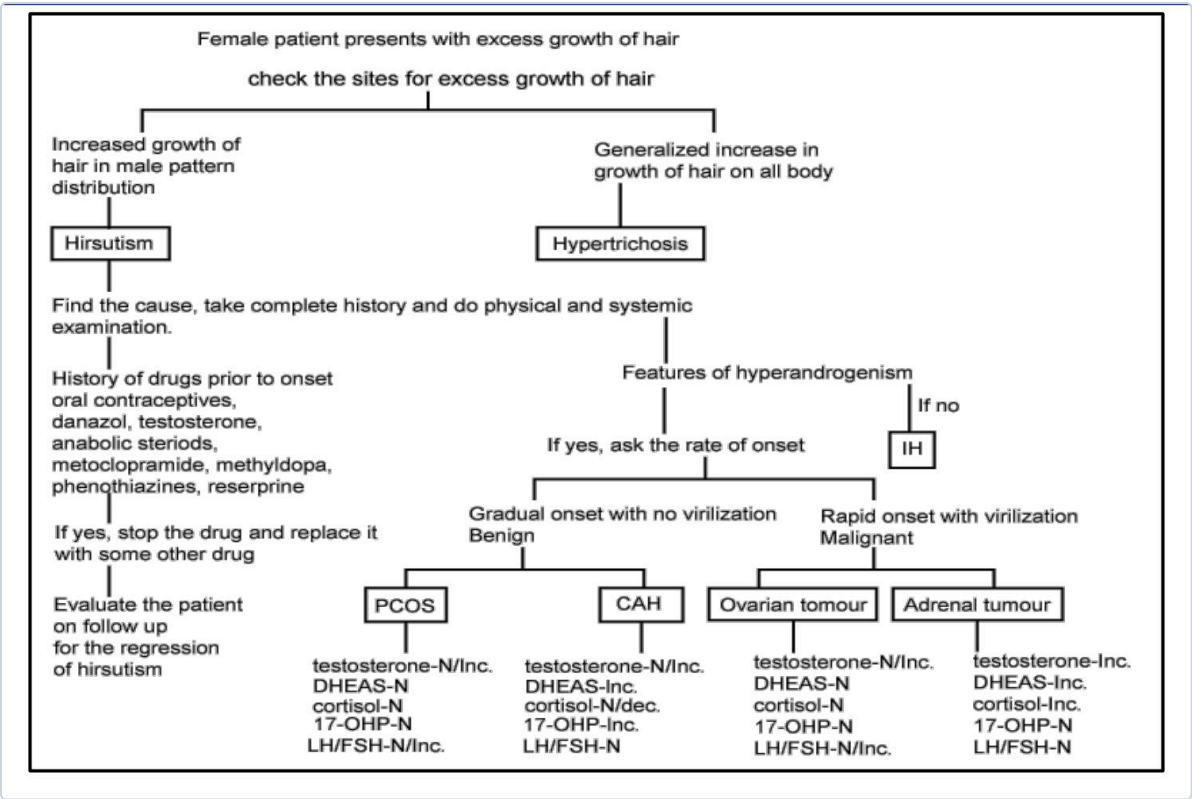
Causes (slide 44):

Androgen-independent (hypertrichosis)	Androgen-dependent (hirsutism)
Constitutional, familial, racial	**Ovarian:** neoplastic (**arrhenoblastoma** = Sertoli–Leydig); non-neoplastic: **idiopathic hirsutism**, ⭐ **PCOS** (*[added]* commonest cause)
Starvation, anorexia nervosa	**Adrenal:** neoplastic (**virilising adrenal tumour**); non-neoplastic: **CAH**, **Cushing's**
Drugs: minoxidil, phenytoin, diazoxide (*[added]* ciclosporin)	**Drugs: androgens** (*[added]* danazol, anabolic steroids)

Diagnostic approach (slide 47) ⭐

1. **Male-pattern** distribution = **hirsutism**; **generalised** = **hypertrichosis**.
2. **History of drugs before onset?** (OCPs, **danazol, testosterone, anabolic steroids, metoclopramide, methyldopa, phenothiazines, reserpine**) → stop or replace the drug, follow up for regression.
3. **No features of hyperandrogenism** → **idiopathic hirsutism (IH)**.
4. Hyperandrogenism → **rate of onset**: - **Gradual, no virilisation = benign** → **PCOS** or **CAH** - ⭐ **Rapid onset with virilisation = malignant** → **ovarian** or **adrenal tumour**

	Testosterone	DHEA-S	Cortisol	17-OHP	LH/FSH
PCOS	N/↑	N	N	N	**N/↑** (*[added]* LH:FSH > 2)
CAH	N/↑	**↑**	N/↓	⭐ **↑**	N
Ovarian tumour	⭐ **N/↑ (often > 200 ng/dL)**	**N**	N	N	N/↑
Adrenal tumour	↑	⭐ **↑**	**↑**	N	N



Slide 47: diagnostic approach to hirsutism

> **Rule** *[added]*: **testosterone very high + normal DHEA-S → ovary**; **DHEA-S very high → adrenal**. **↑ 17-OHP → CAH.**

Treatment (slides 45–46):

1. **Treat the cause:** stop the drug, remove tumours, treat medical disorders. ⭐ **Weight reduction = 1st-line for PCOS** in overweight women.
2. **Target organ:** - **androgen-receptor competition** at follicles: **spironolactone, flutamide, ketoconazole, cyproterone acetate** - **5α-reductase inhibitor**: **finasteride** - *[added]* combined OCP is first-line systemic therapy
3. **Recent: eflornithine (Vaniqa 13.9% cream):** irreversibly inhibits **ornithine decarboxylase** (rate-limiting step of follicular polyamine synthesis). Improvement over **4–8 weeks**; minor skin irritation.
4. **Cosmetic:** - **Bleaching:** irritation, pruritus, discolouration. - ⭐ **Shaving does NOT worsen hirsutism.** It is a good short-term option for facial hair: no effect on anagen rate or duration or on hair diameter; the blunt tip only *looks* thicker. - **Plucking/waxing:** scarring, folliculitis, hyperpigmentation. - **Depilatory creams:** irritant dermatitis. - **Electrolysis:** painful, erythema, scarring. - **Laser.**

> 🔁 **Recap: Hypogonadism and hirsutism.**
>
> - **Hypogonadism:** ↓LH/FSH = pituitary; ↑LH/FSH + ↓T = testicular (Klinefelter); normal/↑ all = resistance. Testosterone IM, not oral; pulsatile LHRH for hypothalamic causes.
> - **Hirsutism** = male-pattern hair in women (androgen-dependent); hypertrichosis = generalised (drugs: minoxidil, phenytoin, diazoxide).
> - **Causes:** PCOS, idiopathic, CAH (↑17-OHP), Cushing's, tumours (rapid + virilisation; ovary ↑T, adrenal ↑DHEA-S).
> - **Treatment:** weight loss (PCOS), anti-androgens (spironolactone, cyproterone, flutamide), finasteride, eflornithine, cosmetic (shaving does not worsen it).

3. High-Yield Comparison Tables

Table A: Primary vs Secondary vs Tertiary Adrenal Insufficiency

	Primary (Addison's)	Secondary (pituitary)	Tertiary (hypothalamic)
Commonest cause	**Autoimmune 80%** (TB 10–20%)	Pituitary tumour/Sheehan	⭐ **Steroid withdrawal**
ACTH	**↑**	↓/N	↓/N
Pigmentation	⭐ **Yes**	**No** (pale)	No
Aldosterone	**↓ → hyperK, salt craving, NAGMA**	**Normal**	Normal
ACTH stimulation	**No rise**	**Rise** (unless long-standing)	None/mild
Fludrocortisone	**Yes**	No	No

Table B: CAH Types

Enzyme	BP	K^+	Androgens	Female	Male	Marker *[added]*
21-OH (commonest)	**↓ (salt-losing)**	**↑**	**↑**	**Virilised** (ambiguous)	**Precocious puberty**	**↑ 17-OHP**
11β-OH (~10%)	**↑**	↓	**↑**	Virilised	Precocious	↑ 11-deoxycortisol, ↑ DOC
17α-OH (rare)	**↑**	**↓**	**↓**	**Sexual infantilism**	**Undervirilised** (pseudohermaphrodite)	↑ DOC, ↓ 17-OHP

Table C: Adrenal Crisis vs Pheochromocytoma Crisis

	Adrenal crisis	Pheo crisis
BP	**Shock** (refractory hypotension)	**Severe paroxysmal HTN** (± postural drop)
Glucose	**↓**	**↑**
Na/K	**↓Na, ↑K**	N
Skin	Pigmented (primary)	**Pale**, sweaty
Treatment	**Hydrocortisone 100 mg IV + saline + dextrose**	**Phentolamine / nitroprusside**, then α-blocker; **no β-blocker first**

Table D: Hirsutism Causes by Labs (Slide 47)

Cause	Onset	Key lab
PCOS	Gradual	N/↑ T, ↑ LH/FSH
CAH	Gradual	**↑ 17-OHP**, ↑ DHEA-S
Ovarian tumour	**Rapid + virilisation**	**↑ T**, normal DHEA-S
Adrenal tumour	**Rapid + virilisation**	**↑ DHEA-S**, ↑ cortisol
Idiopathic	Gradual	All normal
Drugs	After the drug	—

4. Rules, Exceptions, Patterns

Rule	Exception / nuance
Adrenal insufficiency → pigmentation	**Only primary** (↑ ACTH/MSH); secondary/tertiary are **pale**
AI → hyperkalaemia	**Only primary** (aldosterone lost); secondary keeps the RAAS
Confirm the diagnosis first	⭐ **Crisis: treat immediately** with hydrocortisone; test later
Replace thyroid and adrenal	⭐ **Steroid before thyroxine** (Sheehan: thyroxine alone → crisis)
Fludrocortisone in AI	**Primary only**; in crisis, add it when hydrocortisone is ≤ 100 mg/day
Women lose body hair in AI	Men don't (testicular androgen)
CAH → virilisation	**17α** → **no** androgens (infantilism)
CAH → salt loss	**11β and 17α → hypertension** (DOC accumulates)
β-blocker for tachycardia	⭐ **Never first in pheo**: α-blockade first
Paroxysmal HTN = pheo	Only **50%** are paroxysmal; 50% sustained; some normotensive
VMA raised = pheo	**False positive**: vanilla, chocolate, aspirin
Shaving worsens hair	**No**: it only looks thicker
Hirsutism = hypertrichosis	Hirsutism is **androgen-dependent, male-pattern, women only**

5. Clinical Correlations

- **Thin, tired woman with tanned skin, dark buccal patches, BP 90/60, Na 128, K 5.8, vitiligo and hypothyroidism** → **autoimmune Addison's (Schmidt/APS-2)** → ACTH stimulation → hydrocortisone + fludrocortisone.
- **Patient on long-term prednisolone who stops abruptly before surgery and becomes hypotensive** → **tertiary AI / crisis** → IV hydrocortisone.
- **Child with meningococcal purpura and refractory shock** → **Waterhouse–Friderichsen**.
- **Sheehan patient started on thyroxine alone who collapses** → precipitated crisis.
- **Newborn girl with ambiguous genitalia; newborn boy vomiting with K 7.2 at 2 weeks** → **21-OH CAH (salt-wasting)** → ↑ 17-OHP.
- **Hypertensive child with virilisation** → **11β-OH CAH**.
- **Hypertensive 16-year-old girl with absent puberty and low K^+** → **17α-OH CAH**.
- **Young woman with episodic headache, sweating, palpitations, a thyroid nodule and high calcitonin** → **MEN 2A** → metanephrines **before** thyroid surgery (lecture case).
- **Hypertensive given propranolol, then BP rockets** → undiagnosed **pheo** (unopposed α).
- **Woman with rapid-onset hirsutism, clitoromegaly, deep voice, testosterone 300 ng/dL** → **ovarian (Sertoli–Leydig) tumour**.

6. Rapid Revision (Last-Day)

1. Primary AI: **autoimmune 80%** (± Schmidt: T1DM, hypothyroid, pernicious anaemia), **TB 10–20%**, adrenalectomy, infiltration. Secondary: hypopituitarism. **Tertiary: steroid withdrawal (commonest overall).**
2. Weakness, weight loss, **postural hypotension (SBP ≥ 110 excludes)**, hypoglycaemia, **pigmentation (palmar creases, buccal mucosa, scars, nipples; primary only)**, ↓ female axillary and pubic hair, GI symptoms, polyuria, **salt craving**.

3. Labs: **↓Na ↓Cl ↑K acidosis ↑Ca ↓glucose**; eosinophilia, lymphocytosis, normocytic anaemia.
4. **Morning ACTH:** ↑ primary, ↓/N secondary. **ACTH stimulation = gold standard:** normal ↑ ≥ 6 µg/dL; **primary no rise**, secondary rise, tertiary/long-standing little.
5. Cause: adrenal antibodies; adrenal calcification (TB).
6. Treatment: **prednisolone 5 mg am + 2.5 mg pm + fludrocortisone 0.1 mg (primary)**; minor illness → **double**; major surgery → **hydrocortisone 100 mg IV night before, then 100 mg every 8 h, taper over 3–5 days**.
7. **Crisis triggers:** illness without dose increase, **steroid withdrawal**, **Sheehan on thyroxine alone**, adrenalectomy, **Waterhouse–Friderichsen**, metyrapone/mitotane, adrenal vein thrombosis.
8. Crisis: weakness, confusion, **abdominal pain**, vomiting, **shock** → death within 24 h. Suspect in **fluid/pressor-refractory hypotension**.
9. **Hydrocortisone 100 mg IV immediately** (don't wait), then 100 mg every 8 h; **saline + glucose**; fludrocortisone when hydrocortisone ≤ 100 mg/day; treat the cause. **5 S's.**
10. **CAH:** AR; ↓ cortisol → ↑ ACTH → hyperplasia. **21-OH** (commonest): salt-losing + virilisation. **11β** (10%): **HTN** + virilisation. **17α** (rare): **HTN + hypoK** + sexual infantilism / male pseudohermaphroditism.
11. CAH treatment: **prednisolone 7.5 mg/day** (↓ ACTH) ± **fludrocortisone 0.1 mg**; surgery for ambiguous genitalia; lifelong, with stress dosing.
12. **Pheo:** rule of 10s; right side more often; **MEN 2A (Sipple)**; noradrenaline from all extra-adrenal tumours, adrenaline from adrenal ones.
13. **5 P's**; cold pale extremities; HTN **50% paroxysmal / 50% sustained**; postural hypotension; hyperglycaemia; triggers: tumour pressure, exercise, cheese/alcohol, **β-blockers**, labour.
14. Tests: **urine and plasma metanephrines (most sensitive)**, VMA (false +ve with vanilla, chocolate, aspirin), **clonidine suppression** (normal falls; pheo doesn't), CT/MRI/isotope scan, caval sampling.
15. Treatment: **α first** (**phenoxybenzamine**; acute **phentolamine 2–5 mg IV** or nitroprusside) → **β after** (propranolol) → **metyrosine** if inoperable → **surgery**.
16. **Hypogonadism:** ↓LH/FSH = pituitary; ↑LH/FSH ↓T = testicular (Klinefelter, CF, myotonic dystrophy, irradiation); testosterone **IM** (not oral).
17. **Hirsutism** = male-pattern hair in women. Causes: **PCOS**, idiopathic, CAH, Cushing's, tumours (rapid + virilisation), drugs. Hypertrichosis drugs: **minoxidil, phenytoin, diazoxide**.
18. Treatment: weight loss (PCOS), spironolactone/flutamide/cyproterone/ketoconazole, finasteride, **eflornithine** (ODC inhibitor), cosmetic (**shaving doesn't worsen it**).

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7. Exam-Style Questions

Q1 (lecture case, slides 48–49). A 30-year-old woman has an enlarging painful neck swelling, episodic throbbing headaches, sweating and palpitations, HR 112, BP 140/90, a 3-cm thyroid nodule, normal TSH, and **calcitonin 300 ng/dL** (N < 5). Which lab abnormality is most likely? a) ↑ Serum gastrin · b) ↑ Serum cortisol · c) ↑ Urinary 5-HIAA · d) ↑ Serum T3 · **e) ↑ Plasma metanephrines**

Answer: e. *[reasoning added; key not on slides]*

- **High calcitonin + thyroid nodule = medullary thyroid carcinoma.**
- With **episodic headache, sweating and palpitations** → **pheochromocytoma** → **MEN 2A (Sipple)** (slide 28).
- Wrong options:
  - gastrin = MEN 1 / ZES
  - 5-HIAA = carcinoid
  - T3 = thyrotoxicosis (TSH is normal)
  - cortisol = Cushing's
- *[added]* **Remove the pheo first** (after α-blockade), then the thyroid.

Q2. A 45-year-old woman has fatigue, weight loss, postural dizziness and **pigmented palmar creases and buccal mucosa**. Na 127, K 5.9, glucose 3.2 mmol/L. Morning cortisol is low and **ACTH is high**; cortisol does not rise after cosyntropin. Treatment? A. Hydrocortisone alone · **B. Glucocorticoid + fludrocortisone** · C. Thyroxine first · D. Fludrocortisone alone · E. Bromocriptine

Answer: B.

- **Primary AI** (pigmentation, hyperkalaemia, high ACTH, no response) needs **both** glucocorticoid **and mineralocorticoid** replacement (slide 17).
- A would suit secondary AI.

Q3. A known Addison's patient arrives with vomiting, abdominal pain, confusion and BP 70/40 after 3 days of gastroenteritis. First step? A. Await cortisol result · B. ACTH stimulation test · **C. Hydrocortisone 100 mg IV + IV saline and dextrose immediately** · D. Fludrocortisone 0.1 mg PO · E. Dopamine infusion

Answer: C.

- **Adrenal crisis**: treat immediately **without waiting for confirmation** (slide 21).
- Fludrocortisone is added later, once hydrocortisone is ≤ 100 mg/day.

Q4. A newborn girl has ambiguous genitalia, and on day 10 develops vomiting, hypotension, **Na 124 and K 7.0**. Which enzyme is deficient and what is the diagnostic test? A. 11β-hydroxylase; 11-deoxycortisol · **B. 21-hydroxylase; 17-hydroxyprogesterone** · C. 17α-hydroxylase; DOC · D. Aromatase; oestradiol · E. 5α-reductase; DHT

Answer: B.

- **Salt-wasting + virilisation** = **21-OH deficiency** (slide 25). *[added]* ↑ 17-OHP.
- 11β gives **hypertension**.

Q5. A patient with paroxysmal hypertension and raised plasma metanephrines is to have surgery. Which pre-operative regimen is correct? A. Propranolol, then phenoxybenzamine · **B. Phenoxybenzamine, then propranolol once α-blocked** · C. Propranolol alone · D. Hydralazine alone · E. No preparation needed

Answer: B.

- **α-blockers ahead, β-blockers behind** (slide 35). A β-blocker first → unopposed α → hypertensive crisis.

Q6. A 26-year-old woman has 3 months of rapidly worsening facial hair, clitoromegaly and a deepening voice. Testosterone is markedly raised and DHEA-S is normal. Likely source? A. PCOS · B. 21-OH CAH · **C. Ovarian androgen-secreting tumour** · D. Adrenal carcinoma · E. Idiopathic

> **Answer: C.**
> - **Rapid + virilisation = malignant/tumour**.
> - **High testosterone with normal DHEA-S → ovary** (slide 47).
> - An adrenal tumour would raise **DHEA-S**.

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8. Exam Pitfalls & Traps

- **Pigmentation and hyperkalaemia = PRIMARY AI only.** Secondary/tertiary are pale with normal K^+ (link to Simmonds disease).
- **SBP ≥ 110 "excludes Addison's"** (slide 7). That is the lecture's rule of thumb; don't over-apply it in a compensated patient.
- **ACTH stimulation threshold:** the slide says a rise of **≥ 6 µg/dL above basal**. Modern guidelines use **peak ≥ 18 µg/dL** *[added]*.
- **Fludrocortisone dose:** slide 17 says **"0.1 mg every other day"**, while the crisis and CAH slides say **0.1 mg/day**. Standard maintenance is **0.05–0.2 mg daily** *[added]*.
- **Crisis hydrocortisone:** the lecture gives **100 mg IV, then 100 mg every 8 h**; the 5 S's box says **100 mg then 200 mg/24 h**. Both are ≈ 200–300 mg/day. Never delay the first 100 mg.
- **Sheehan + thyroxine alone → crisis.** Steroid always goes first.
- **CAH 11β vs 21:** both virilise; **11β → HTN**, **21 → salt-wasting hypotension**.
- **17α-OH:** HTN + hypoK + **no sex steroids** (girls don't go through puberty).
- **Pheo: never a β-blocker first**; **never biopsy**; check metanephrines **before** surgery for MTC in MEN 2.
- **VMA false positives:** vanilla, chocolate, aspirin.
- **Glucagon and histamine provocation tests** (slide 32) are **dangerous and obsolete** *[added]*; know them as lecture content only.
- **Hirsutism ≠ hypertrichosis.** Minoxidil, phenytoin and diazoxide cause **hypertrichosis** (androgen-independent).
- **Shaving does not worsen hirsutism.** Slide 46 is explicit.